unification, discussed in the next section: grand unification seeks a single, semi-simple (Lie) group—a group with a single factor—that describes the forces at higher energy scales.

11. The mathematically inclined reader should note that Georgi and Glashow's grand unified theory was based on the group SU(5), which includes SU(3), the group associated with the strong nuclear force, and also SU(2) × U(1), the group associated with the electroweak force. Since then, physicists have studied the implications of other potential grand unified groups, such as SO(10) and E_6.

Chapter 10

1. As we've seen, the big bang's bang is not an explosion that took place at one location in a preexisting spatial expanse, and that's why we've not also asked *where* it banged. The playful description of the big bang's deficiency we've used is due to Alan Guth; see, for example, his *The Inflationary Universe* (Reading, Eng.: Perseus Books, 1997), p. xiii.

2. The term "big bang" is sometimes used to denote the event that happened at time-zero itself, bringing the universe into existence. But since, as we'll discuss in the next chapter, the equations of general relativity break down at time-zero, no one has any understanding of what this event actually was. This omission is what we've meant by saying that the big bang theory leaves out the bang. In this chapter, we are restricting ourselves to realms in which the equations do not break down. Inflationary cosmology makes use of such well-behaved equations to reveal a brief explosive swelling of space that we naturally take to be the bang left out by the big bang theory. Certainly, though, this approach leaves unanswered the question of what happened at the initial moment of the universe's creation—if there actually was such a moment.

3. Abraham Pais, *Subtle Is the Lord* (Oxford: Oxford University Press, 1982), p. 253.

4. For the mathematically inclined reader: Einstein replaced the original equation $G_{\mu\nu} = 8\pi T_{\mu\nu}$ by $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}$ where Λ is a number denoting the size of the cosmological constant.